

Write a program to implement QUEUE using arrays that performs following operations

- (a) INSERT
- (b) DELETE
- (c) DISPLAY

Concept of Queue

A **Queue** is a **linear data structure** that follows the principle **FIFO (First In First Out)**.

- **INSERT** operation is performed at **last**
 - **DELETE** operation is performed at **First**
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Algorithm for Queue using Array

Initialization

- Set `first = -1`
 - Set `last = -1`
 - Declare array `queue [MAX]`
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(a) INSERT Operation

1. If `last == MAX-1`, display **Queue Overflow**
 2. If `first == -1`, set `first = 0`
 3. Increment `last`
 4. Insert element at `queue[last]`
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(b) DELETE Operation

1. If `first == -1` display **Queue Underflow**
 2. Display deleted element `queue[first]`
 3. Increment `first`
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(c) DISPLAY Operation

1. If `first == -1`, display **Queue is Empty**
 2. Print elements from `first` to `last`
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C Program to Implement Queue using Array

```
#include <stdio.h>
#define MAX 5

int queue[MAX];
int first = -1, last = -1;

/* INSERT operation */
void insert()
{
    int item;
    if (last == MAX - 1)
    {
        printf("Queue Overflow\n");
    }
    else
    {
        if (first == -1)
            front = 0;
        printf("Enter element to insert: ");
        scanf("%d", &item);
        last++;
        queue[last] = item;
        printf("Element inserted successfully\n");
    }
}

/* DELETE operation */
void delete()
{
    if (first == -1 || first > last)
    {
        printf("Queue Underflow\n");
    }
    else
    {
        printf("Deleted element: %d\n", queue[first]);
        first++;
    }
}

/* DISPLAY operation */
void display()
{
    int i;
    if (first == -1 || first > last)
    {
        printf("Queue is Empty\n");
    }
    else
```

```

    {
        printf("Queue elements are:\n");
        for (i = first; i <= last; i++)
        {
            printf("%d ", queue[i]);
        }
        printf("\n");
    }
}

int main()
{
    int choice;
    do
    {
        printf("\n--- QUEUE OPERATIONS ---\n");
        printf("1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                insert();
                break;
            case 2:
                delete();
                break;
            case 3:
                display();
                break;
            case 4:
                printf("Exiting program\n");
                break;
            default:
                printf("Invalid choice\n");
        }
    } while (choice != 4);

    return 0;
}

```